BETAVOLTAIC MICROBATTERIES USING POROUS SILICON

Hang Guo[1*], Hui Yang[1] and Ying Zhang[2]
[1]Pen-Tung Sah MEMS Research Center, Xiamen University, Xiamen, Fujian 361005, China
[2]Chemistry and Chemical Biology Department, Cornell University, Ithaca, NY 14853, USA
[*]Tel 86-592-2181340, Fax 86-592-2187196, Email hangguo@xmu.edu.cn

ABSTRACT

In this paper, we present a new type of betavoltaic microbattery using porous silicon *pn* junction device. Investigations are made on using high energy radioisotopes as the source and mechanisms of increased energy bandgap and electron trapping in porous silicon for increasing power output and energy conversion efficiency. The device is fabricated and experimental result shows that this new type of microbattery is a promising technology for developing nuclear microbatteries with the increased power output and conversion efficiency.

INTRODUCTION

MEMS and nanotechnology have gained great progresses during the past 20 years, with various types of devices developed in the size of millimeter to micrometer and even nanometer. To follow this trend, the power sources should be miniaturized in the same pace. Among the various technologies for micro power generation being investigated, nuclear microbatteries that convert the kinetic energy of charged particles emitted from the radioisotopes into electricity are very attractive for many long term applications, such as implanted microdevices for medical treatment and micro-sensors or sensor networks for environmental monitoring. The radioisotopes have the energy density 10^2~10^4 times greater than fossil or chemical fuels, and nuclear microbatteries with a long lifetime can be realized if a proper radioisotope is selected. As one type of nuclear microbatteries developed, betavoltaic microbatteries, in which beta particles emitted from the radioisotopes enter the *pn* junction device to generate electron-hole pairs (EHPs) to directly generate electricity, have been proved an effective technology to provide on-chip micropower for MEMS and nano devices [1]. Due to the low activity utilized, the output power of the silicon betavoltaic microbatteries is currently in the scale of nanoWatt, with the energy conversion efficiency of 0.01~0.5%.

Two basic types of method can be used for increasing the power output and energy conversion efficiency. One type is directly using high energy beta source. The other type is to improve or develop new type of *pn* junction device to further increase the conversion efficiency. Some groups are studying using wide bandgap semiconductors, including SiC [2] or GaN [3], to raise output power, especially the open circuit voltage. However, this requires special facilities for processing. Wen Sun et al [4] recently published a paper about the porous silicon diode for betavoltaics, where silicon pores are used to storage tritium. However, gaseous tritium can be easily inhaled by the people in operation, and ability of silicon pores to storage gaseous tritium is very limited. Thus, the energy conversion efficiency of their device is 0.22%.

In our current study, we tend to improve betavoltaic microbatteries from both radioisotopes adopted and the *pn* junction device fabricated. In this paper, we present a betavoltaic microbattery in which the silicon pores is used for increasing power output and energy conversion efficiency, instead of storing the radioactive source. The radioisotope Pm-147, which has high energy than the radioisotope Ni-63 used before, is selected as the source. Mechanism of increased energy bandgap in porous silicon is discussed. Based on this, the device is fabricated and tested.

SELECTION OF RADIOISOTOPES

It is necessary to discuss selection of radiation source for betavoltaic microbatteries at first, since it is a critical aspect in developing technologies. In general, selection of the radioisotope is based on the types of radiation, safety, energy, specific activity, cost, and half life, and safety is always a great concern when using radioisotopes. Pure beta emitters are best suited for nuclear microbatteries. Table 1 shows the pure beta source candidates that could be used for this application. The radioisotope Ni-63 is very attractive as its lifetime is over 100 years, which means for most of practical applications it could be used forever. And, Ni-63 can be provided in the liquid solution of $Ni^{63}Cl$ mixed with HCl, which allows it to be poured into various micromachined three dimensional *pn* junction devices. However, the kinetic energy carried by the electrons emitted from Ni-63 is low, with the average energy of 16.7keV and highest energy of 67keV, which makes it difficult to

obtain high efficiency microbatteries. The radioisotope Pm-147 becomes our choice since the average kinetic energy of electrons emitted from Pm-147 is 62keV, and highest energy is 250keV, which has been turned out to be safe for both silicon devices and people in handling it [5]. Thus, it does not need to specially design and fabricate a *pn* junction device to avoid the crystal damage caused by the energetic charged electrons. Though the half-life time of Pm-147 is 2.7 years, it is still very long compared with other technologies developed for micro power generation. For example, micro fuel cells can only work for several hours.

Table 1 Pure beta sources for nuclear microbatteries

Radioisotope	Specific Activity (g/mCi)	Half-life Time	Average Energy (keV)
Nickel-63	1.8×10^{-5}	100.2 years	17.4
Tritium-3	1.03×10^{-7}	12.3 years	5.7
Promethium-147	1.06×10^{-6}	2.6 years	62
Stronium-90	7.25×10^{-6}	28.8 years	195.8
Krypton-85	2.56×10^{-6}	10.8 years	251.6
Ruthium-106	3.03×10^{-7}	1.06 years	400
Calcium-45	5.06×10^{-8}	162 days	256
Sulfur-35	2.4×10^{-8}	87.2 days	167

Fig.1 shows the schematic diagram of betavoltic microbattery. From its current model, its open circuit voltage V_{oc} and short circuit current I_s can be derived as[1]

$$V_{oc} = \frac{kT}{e}\ln\left(1+\frac{I_p}{I_0}\right) \quad (1)$$

$$I_s = I_p \quad (2)$$

where I_p is the current generated by the radioisotope and I_0 is the leakage current of *pn* junction device, and maximum output power can be derived as

$$P_{\max} = fV_{oc}I_s \quad (3)$$

where f is the filling factor of *pn* junction device.

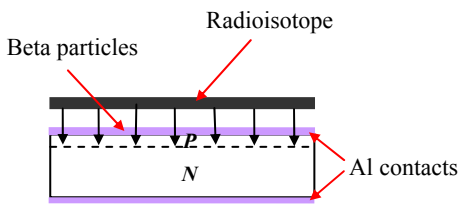


Fig.1 **Schematic diagram of betavoltaic microbattery**

In silicon, it will consume about 3.3eV to generate an EHP [6]. So, a beta particle emitted from Ni-63 with average kinetic energy of 16.7 keV can theoretically generate about 5,000 EHPs. As a comparison, a beta particle emitted from Pm-147, with average energy of 62keV, can theoretically generate about 19,000 EHPs. With an increased short circuit current I_s or I_p, the ratio of (I_p/I_0) is greatly increased, which leads to the increase of open circuit voltage as well, according to Eq.(1). Thus, the utilization of Pm-147 will greatly increase power output of the microbattery, compared with the one using Ni-63. High energy radioactive source, if safely allowed for silicon device and people to handle it, is recommended for developing nuclear microbatteries. When the same amount of radioactivity is used, the microbattery using Pm-147 will output at least 3 times of power larger than the one using Ni-63.

POROUS SILICON *PN* JUNCTION DEVICE

Porous silicon (PS) has been extensively studied during the past two decades. Many applications, such as PS solar cell and PS as a new type of sacrificial layer material for MEMS devices, are being exploited. As mentioned in the previous section, a betavoltic microbattery using porous silicon *pn* junction device was developed by Wen Sun et al, and we schematically plot their structure in Fig.2. In their device, after silicon pores are formed chemically, diffusion of dopant into the porous silicon is done to create the *pn* junction over the entire surface of the device. The working area of this type of device has been greatly increased, but the low ability of pores to store gaseous tritium leads to very low radioactivity utilized in a unit of working area. As a result, it is difficult to obtain a large power output from the microbattery.

Fig.2 **Schematic diagram of the betavoltaic microbattery where the silicon pores is used for storing gaseous tritium**

We proposed and developed another new type of betavoltaic microbattery using porous silicon, where the nanometer pores are formed on the surface of a pn junction device, shown in Fig. 3.

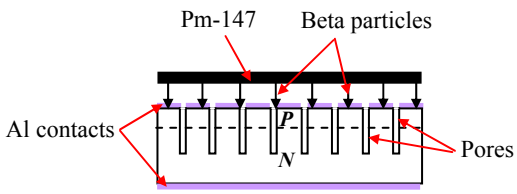
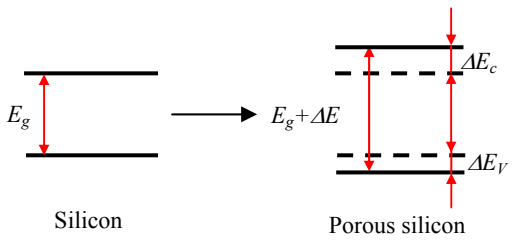


Fig.3 **Schematic of proposed new type of betavoltaic microbattery using porous silicon**

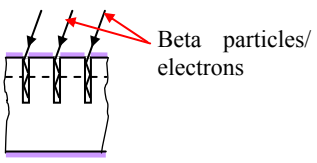
Since Canham [7] reported the strong visible photoluminescence from porous silicon in 1990, considerable studies have been made in its theory and applications. Since the diameter of pores is in the scale of nanometers, the pores can be considered as

one dimensional quantum wells for electrons [8]. As a result, the energy bandwidth of porous silicon has been increased, which was verified by the experiments on its peak shift of photoluminescence. It is regarded by many investigators that the energy bandwidth of silicon with nanometer pores is about 2eV[9]. According to Olsen's study [10], semiconductors with bandgap of ~2eV can own the theoretical energy conversion efficiency of 10~15% for betavoltaic batteries, compared with 3~6% for silicon that own the energy bandwidth of only 1.12eV. Currently, the size of silicon pores fabricated by different research groups can range from several nanometers to several hundreds of nanomters. The mechanism of increasing energy bandgap may not appropriate for all cases. Our point of view is that the energy bandgap of the porous silicon can be greatly affected by the diameter of pores, and if the diameter of the pores is in the scale of several nanometers or even smaller, quantum effect may dominate and energy bandgap may increase sharply; if the diameter of the pores is in the scale of several tens of nanometers, the effect of energy bandgap increased may not exist but the pores can be regarded as the macroscopic wells to trap the electrons emitted from the radioisotope, greatly reducing their bounce-back. Thus, the power output and conversion efficiency can be increased as well.

Fig. 4 shows two mechanisms for our proposed betavoltic microbattery using porous silicon, in which both microscopic and macroscopic effects of silicon pores may be used to increase the power output and energy conversion efficiency, instead of just using the 'geometrical' effect of silicon pores to increase the working area and store the gaseous radioisotopes.



(a) Energy bandgap of silicon and porous silicon



(b) Electron trapping in pores with diameter of several tens of nanometer

Fig.4 **Explanation of two mechanisms in proposed betavoltaic microbattery using porous silicon**

In practice we found that the pores in the scale of nanometer formed by using the electrochemistry method are more stable and easily controlled than the ones using the chemical method. Solid-source diffusion in porous silicon has not been thoroughly investigated. Therefore, instead of the formation of porous silicon layer first, the planar *pn* junction device is created first by diffusing boron into the n-type silicon wafer, and then the electrochemistry method is used to form the porous silicon layer. This microfabrication procedure can ensure the availability of the desired devices, and parameters needed to be adjusted are only for electrochemistry. The junction depth is 2μm, and porous silicon layer is 3μm deep, penetrating through the depletion region of the *pn* junction. Aluminum is sputtered on both sides of the chip as electrodes.

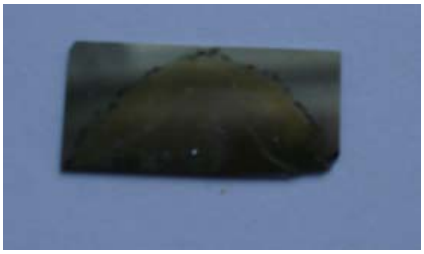


Fig.5 **Picture of porous silicon after electrochemistry.**

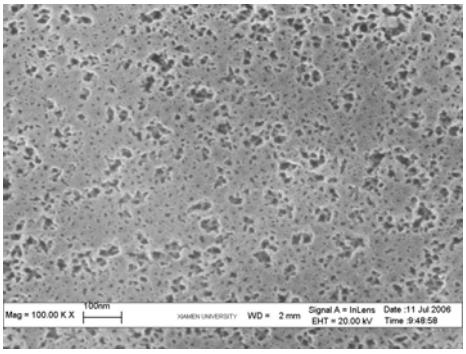


Fig.6 **SEM picture of porous silicon (without aluminum layer)**

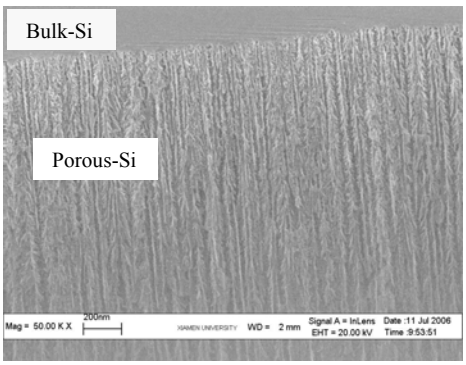


Fig.7 **SEM picture of cross-section of porous silicon**

Fig.5 shows the picture of porous silicon after electrochemistry. Actually, the brown area of silicon piece in the picture is electrochemically etched into

porous silicon, from which an area of 3mm×3mm is cut for further metallization. Fig.6 shows the SEM picture of the surface of porous silicon and Fig.7 shows the SEM picture of its cross-section. It is noted that the silicon pores are not uniform in size. Most of them are 10~20nm in diameter, but some of them are around 100nm, and for a silicon pore, it has many tiny branches.

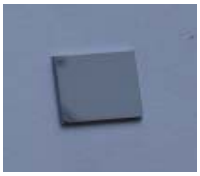


Fig.8 **Porous silicon pn junction device for betavoltaic microbattery**

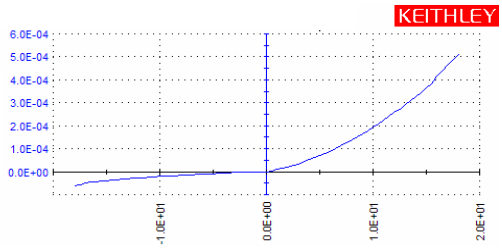


Fig.9 **I-V curve of porous silicon *pn* junction device**

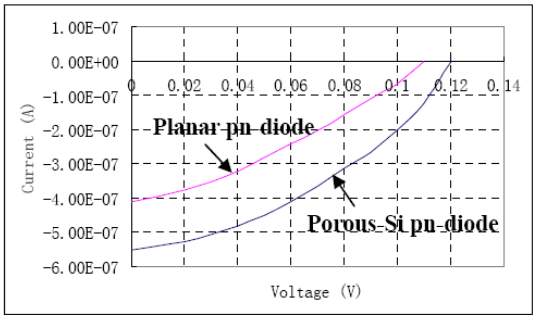


Fig.10 **Tested result of betavoltaic microbattery using porous silicon and Pm-147**

Fig.8 shows the porous silicon *pn* junction device after metallization. We test its *I-V* curve as shown in Fig.9. Its leakage current is 1.7*pA*. As a betavoltaic microbattery, a piece of 8mCi Pm-147 source plate is used to cover the fabricated device for the experiment, and result is shown in Fig. 10.

As shown in Fig.10, the microbattery using the porous silicon *pn* junction device has the short circuit current of 0.55μA, open circuit voltage of 0.12V, and maximum output power of about 0.044μW. As a comparison, microbattery using a planar *pn* junction device with the same geometrical and diffusion parameters is also tested with the same radioactive source. The microbattery using planar *pn* junction device has the short circuit current of 0.41μA, open circuit voltage of 0.11V, and maximum output power of about 0.031μW. Compared with the microbattery using planar *pn* junction device, the output power of microbatttery using porous silicon raise 42%, and the energy conversion efficiency of this battery using porous silicon and Pm-147 is 1.75%. We think that two mechanisms discussed above may all exist in this device, leading to the increased power output and energy conversion efficiency. More theoretical and experimental work are being done to further increase the power output and energy conversion efficiency of the betavoltaic microbattery using porous silicon and high energy solid state beta sources.

CONCLUSION

In this paper, we proposed and fabricated a new type of betavoltaic microbattery using porous silicon *pn* junction device and the radioisotope Pm-147. The experimental result shows that the low cost porous silicon junction device fabricated by using electrochemistry method can be a new converter for betavoltaic microbatteries.

REFERENCES

[1] Hang Guo, Amit Lal, Nanopower betavoltaic microbatteries, Proceedings of Transducers'03, 33-36

[2] M.V.S Chandrashekhar, C.I.Thomas, H.Li, M.G.Spencer, A. Lal, Demonstration SiC betavoltaic cell, Applied Physics Letters, 88, 2006, pp.033506-9

[3] Christiana Honsberg, William Doolittle, Mark Allen, GaN betavoltaic energy converters, Proceedings of the 31st IEEE Photovoltaics Specialist Conference, Orlando Florida, 2005

[4] Wen Sun, Nazir Kherani, Karl Hirschman, Larry Gadeken, Philippe Fauchet, A three-dimensional porous silcion p-n diode for betavoltics and photovotaics, Advanced Materials, 2005, 17, pp.1230-1233

[5] Hang Guo, et al, Nuclear microbatteries for MEMS and nano devices, Proceedings of APCOT 2006, June, 2006, Singapore, 95-PMEM-A0393

[6] C.A. Klein, Bandgap dependence and related features of radiation ionization energies in semiconductors, Journal of Applied Physics, Vol.39, No.4, March, 1968, pp.2029-2038

[7]L. Canham, Silicon quantum wire array fabrication by electrochemical and chemical dissolution of wafers, Applied Physics Letter, 1990, 57, pp.1064

[8]M. S. Brandt, et al, The origin of visible luminescencefrom "porous silicon": A new interpretation, Solid State Communications, Vol.81, pp. 307-312

[9].P.Vitanov, et al, High-efficiency solar cell using a thin porous silicon layer, Thin Solid Films, 1997, pp.299-303

[10]. L.C Olsen, Advanced betavoltaic power source, Proceedings of 9th Intersociety Energy Conversion Engineering Conference, 1974, pp.754